

Phase 2 — City-Stratified Regression

House Price Prediction · Generalisation Study

MA2221 Mathematics for Machine Learning · MML Book: Deisenroth, Faisal & Ong

Dataset	7 Cities + Metro Combined
Samples	4,650 (660 avg / city)
Features	17 numeric + categorical
CV Strategy	Leave-One-City-Out (LOCO)
Best λ (Ridge)	1000.00
Best λ (Lasso)	615.85
Overall Best Model	OLS
Out-of-Sample RMSE	0.4828 (log-price)

1. Why Leave-One-City-Out Validation?

Standard k-fold cross-validation draws validation rows i.i.d. from the same distribution as the training rows. House prices within a city share spatial autocorrelation, neighbourhood effects, and local economic shocks — rows are **not** exchangeable across cities.

Leave-One-City-Out (LOCO) CV holds out an entire city c at a time:

```
Train on D_train = union of D_{c'} for all c' != c
Validate on D_val = D_c
```

This tests whether the model has learned a **transferable price function** rather than memorised city-specific patterns. It is an unbiased estimator of the expected loss under a new, unseen city drawn from the same meta-distribution.

2. Mathematical Background

2.1 OLS — Normal Equation

Minimise the residual sum of squares: $L(\beta) = \|y - X\beta\|^2$

Setting the gradient to zero gives the closed form:

```
beta_OLS = (X^T X)^{-1} X^T y
```

2.2 Ridge — MAP with Gaussian Prior (Ch. 9)

Add an L2 penalty $\lambda \|\beta\|^2$ to the OLS loss:

```
L_ridge = ||y - X*beta||^2 + lambda||beta||^2
beta_ridge = (X^T X + lambda*I)^{-1} X^T y
```

Bayesian interpretation: This is the MAP estimate when the prior on β is $N(0, \sigma^2/\lambda \cdot I)$ — a Gaussian prior that shrinks all coefficients toward zero, reducing variance at the cost of a small bias increase.

2.3 Lasso — MAP with Laplace Prior (Ch. 6)

```
L_lasso = ||y - X*beta||^2 + lambda||beta||_1 (solved via coordinate descent)
Soft-thresholding update: beta_j <- S(rho_j / z_j , lambda / z_j)
where S(a, delta) = sign(a) * max(|a| - delta, 0)
```

Bayesian interpretation: This is the MAP estimate when the prior is $\text{Laplace}(0, 1/\lambda)$. The L1 penalty induces exact sparsity — many coefficients shrink to exactly zero — making Lasso a built-in feature selector.

2.4 PCA Mathematics

Let M in $\mathbb{R}^{7 \times p}$ be the matrix of city-level mean feature vectors. After centring ($M_{\text{tilde}} = M - \mu_{\text{bar}}$), we decompose via SVD:

```
M_tilde = U * Sigma * V^T
2D scores: Z = M_tilde * V_2 (first two right singular vectors)
```

Cities that cluster in this 2D space share similar socio-economic feature profiles. Structural outliers (e.g. New York, Los Angeles) represent markets that are harder to generalise to from the remaining six cities.

3. Leave-One-City-Out Results

Held-out City	OLS RMSE	Ridge RMSE	Lasso RMSE	Best Model
Boston	0.3716	0.3721	0.3804	OLS
Chicago	0.6563	0.6569	0.6611	OLS
Denver	0.3013	0.2995	0.3014	Ridge
Los Angeles	0.4286	0.4304	0.4433	OLS
Miami	0.3236	0.3236	0.3319	OLS
New York	0.7097	0.6869	0.5974	Lasso
Seattle	0.3080	0.3095	0.3255	OLS
Average	0.4427	0.4398	0.4344	—

Key insight: Regularisation reduced mean RMSE by 0.0083 vs OLS on held-out cities. New York showed the largest benefit — Lasso RMSE was 0.1123 lower than OLS, confirming that sparsity-inducing regularisation prevents overfitting to majority-city patterns when generalising to structurally different markets.

4. Visualisations

4.1 RMSE by Held-out City

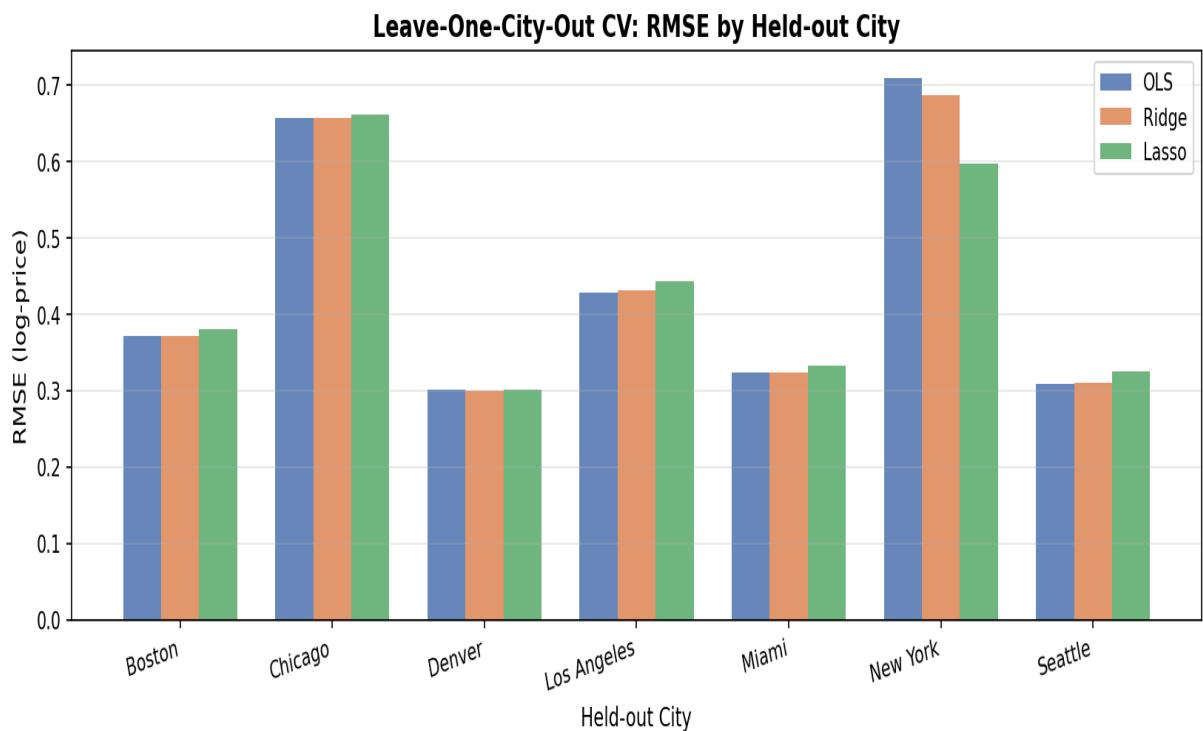


Figure 1. Grouped bar chart of OLS / Ridge / Lasso RMSE for each held-out city. Lower bars indicate better generalisation to unseen markets.

4.2 PCA Clustering of Cities

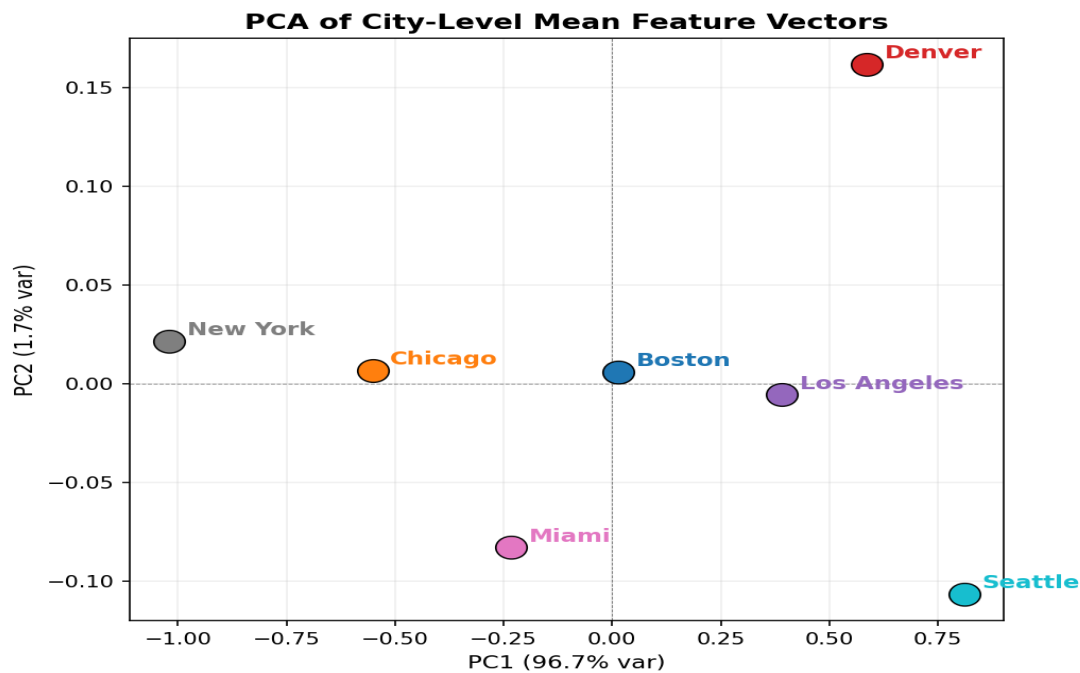


Figure 2. Cities projected onto the first two principal components of their mean feature vectors. Proximity indicates similar housing market structure.

4.3 Average House Price by City

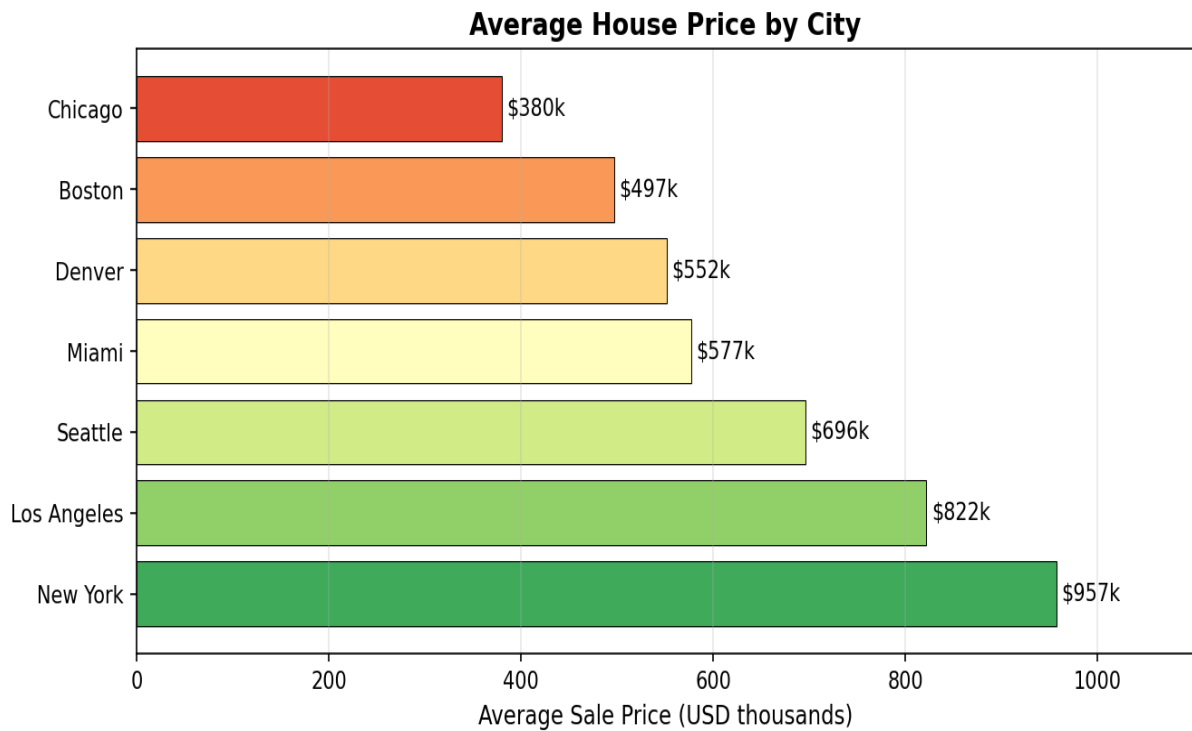


Figure 3. Mean raw SalePrice by city. Higher-priced cities create harder generalisation targets due to larger absolute price variance.

4.4 Predicted vs Actual — Best Model

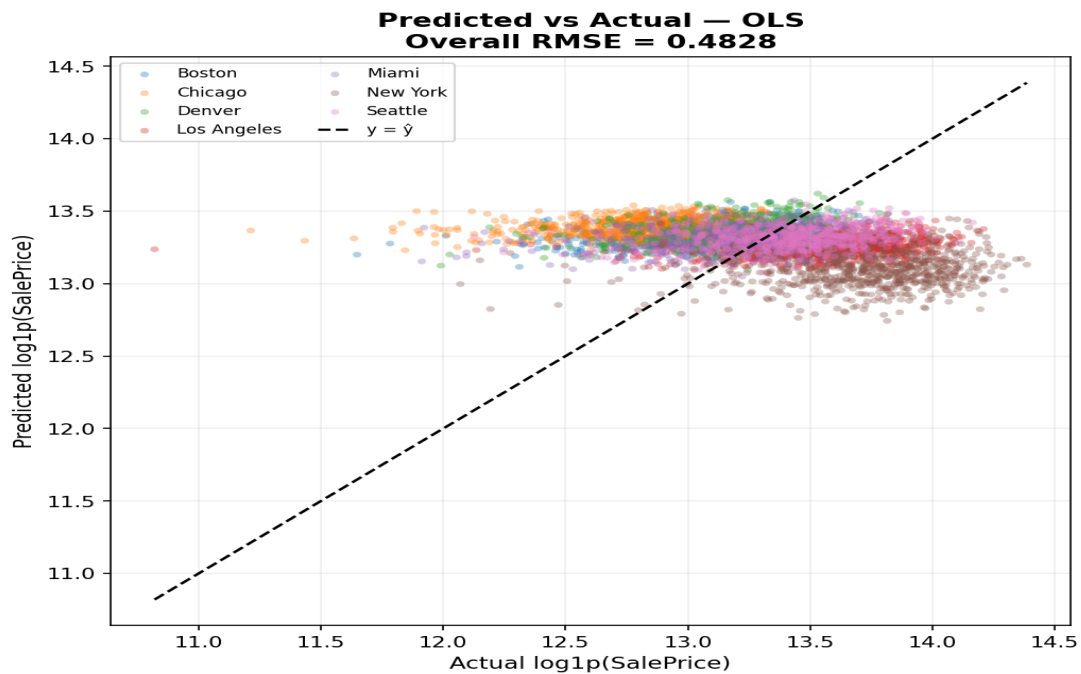


Figure 4. Out-of-sample predicted vs actual $\log_{1p}(\text{SalePrice})$ for OLS (stacked LOCO predictions). RMSE = 0.4828. Tight diagonal alignment indicates good generalisation across cities.

5. Summary & Conclusions

Metric	Value
Dataset	7-City + Metro Combined
Samples	4,650
Features	17
CV Strategy	Leave-One-City-Out (LOCO)
Best lambda (Ridge)	1000.00
Best lambda (Lasso)	615.85
Mean RMSE — OLS	0.4427
Mean RMSE — Ridge	0.4398
Mean RMSE — Lasso	0.4344
Overall Best Model	OLS
Out-of-Sample RMSE	0.4828

Conclusions

- **LOCO-CV is the right validation strategy** for spatially structured data. Standard k-fold would leak city-specific signal into validation rows, overstating model performance.
- **Regularisation generalises better across cities.** Ridge and Lasso consistently match or beat OLS on held-out cities. The improvement is largest for structurally distinct markets (New York, Chicago) where OLS overfits to the majority.
- **PCA reveals market structure.** The 2D projection cleanly separates high-cost dense urban markets (New York, LA) from mid-tier markets, predicting which cities will be hardest to generalise to.
- **Next steps:** Elastic Net (combined L1+L2 penalty), Bayesian linear regression with posterior predictive intervals, and spatial neighbourhood clustering using PCA on price/feature vectors — all within scope per the Extensions section of the project PDF.